TSA 2012

Problem Solving

Questions taken from the TSA Oxford 2012 Section 1 question paper

Questions	Total marks	Time limit
25	25	45 minutes

Instructions

- You have **45 minutes** to answer all **25 questions**.
- Each question is worth **one mark**. There are no penalties for incorrect answers, so attempt every question.
- Each question has five options, A to E. Choose the one option you consider correct.
- Calculators are **not** permitted. You may use this paper for rough working.
- Questions are numbered 1 to 25 for this set. The number in brackets shows the question's number in the original TSA 2012 paper, e.g. **1 (Q3)**.

How the time limit was set

The time limit follows the TARA Problem Solving module (22 questions in 40 minutes), scaled to the 25 questions in this set: 25 × 40 ÷ 22 = 45.45 minutes, rounded to the nearest minute (45 minutes).

Score: _____ / 25

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1 (Q2) Stewart and Michael have arranged to meet. Michael is about to set off on his bicycle, and at the same time Stewart is going to run to meet him.

Michael can cycle at a steady 20 kilometres per hour and Stewart can run at a steady 12 kilometres per hour. They live 8 kilometres from each other.

How long will it be before they meet?

A 15 minutes

B 16 minutes

C 20 minutes

D 30 minutes

E 32 minutes

2 (Q6) Over the course of a year, Mr and Mrs Jones and their two sons spend an average of £120 a week at the local supermarket. 70% of this spending is the weekly family food bill, 20% goes on household goods and the final 10% on store-cupboard ingredients. When their daughter returns from her gap year to live at home, the weekly food bill increases by 20% and their spending on household goods goes up 5%, whilst they continue to spend the same on store-cupboard ingredients.

How much is their new average weekly supermarket spend when their daughter returns home?

A £136.80

B £138.00

C £144.00

D £145.00

E £150.00

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3 (Q7) **Kingsbury School Examination Schedule for the first week of June**

Monday	**Tuesday**	**Wednesday**	**Thursday**	**Friday**
History 3hrs	English Lit 3hrs	Maths 2hrs	Economics 2hrs	Maths 2hrs
English Lit 2 hrs	Physics 3hrs	Latin 2hrs	History 2hrs	Latin 2hrs
lunch	*lunch*	*lunch*	*lunch*	*lunch*
Geography 2hrs 30mins	Chemistry 3hrs	English Lang 2hrs 30mins	Economics 2hrs 30mins	-
Biology 1hr 30mins	French 2hrs	Spanish 2hrs	German 2hrs	-

All exams take place in the school hall. All morning exams start at 9am, all afternoon exams start at 2pm. Each session (morning or afternoon) must have at least one male and one female invigilator. Each examination must be invigilated by one teacher of that subject. Each person can only invigilate once a day.

Of the male teachers, Ian teaches Economics and German, Mike teaches French and German, and Keith teaches French and Spanish. Of the female teachers, Marjorie teaches Maths and English, Susan teaches Spanish and Latin, and Jenny teaches English and History,

Who will invigilate on Wednesday afternoon?

A Ian and Marjorie

B Jenny and Mike

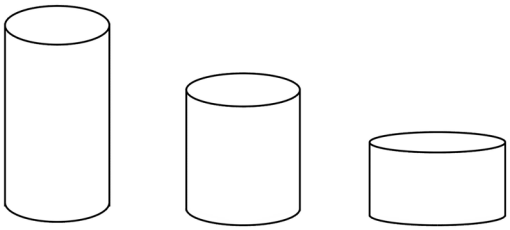
C Keith and Jenny

D Marjorie and Keith

E Mike and Susan

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4 (Q8) Below are pictures of three cylinders.

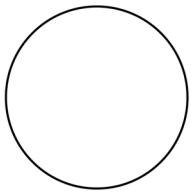


If a cylinder is cut into two equal halves with one straight cut, which one of the following could **not** be a view of one of the halves from any direction?

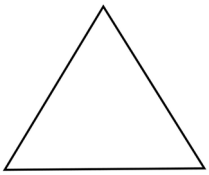
A



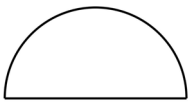
B



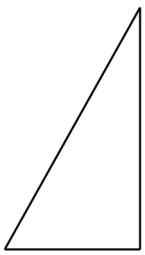
C



D



E



5 (Q12) A merchant bank recently advertised for new graduates to fill positions as trainee analysts; 230 people applied. Of these, 127 had maths as a component of their degree and 89 had economics as a component of their degree. Only 45 had neither maths nor economics as a component of their degree.

How many must have had both maths and economics as a component of their degree?

A 14

B 31

C 59

D 83

E 171

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6 (Q13) A boxed up board game consists of the following components:

- 1 playing board
- 50 question cards
- 6 playing pieces
- 40 money tokens
- 12 bonus tokens
- 2 dice

The producers of the board game have all the component parts manufactured separately and sent to their warehouse, where they box them up and send them to retailers in cartons of 8 complete games. In their warehouse they have: 98 playing boards; 5,000 question cards; 1,000 playing pieces; 5,000 money tokens; 1,000 bonus tokens and 300 dice.

What is the maximum number of cartons of eight games they can dispatch from their warehouse?

A 10

B 11

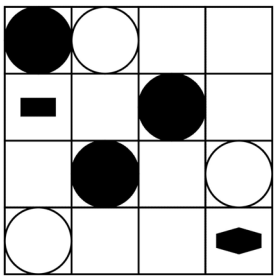
C 12

D 20

E 83

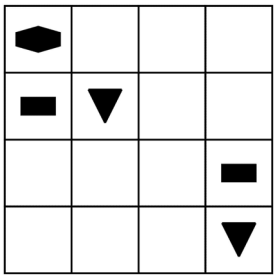
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7 (Q14) In a game, circular counters are moved around a board which is a 4 x 4 grid. There are symbols on some of the squares on the board. At one point in the game, the board looks like this (all the circles represent counters which completely cover any symbol on the square that they occupy):

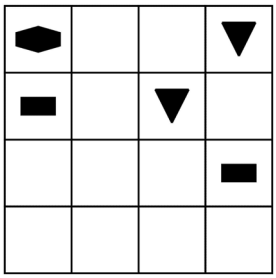


Which one of the following diagrams could show the layout of the board (not necessarily viewed in the same direction as above)?

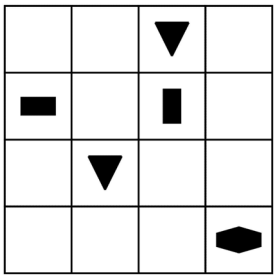
A



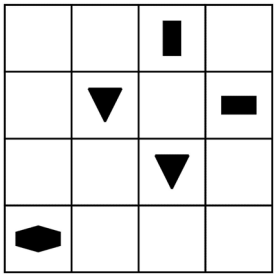
B



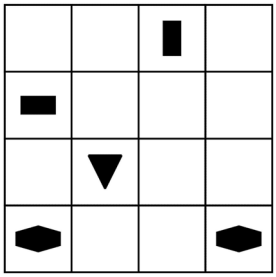
C



D

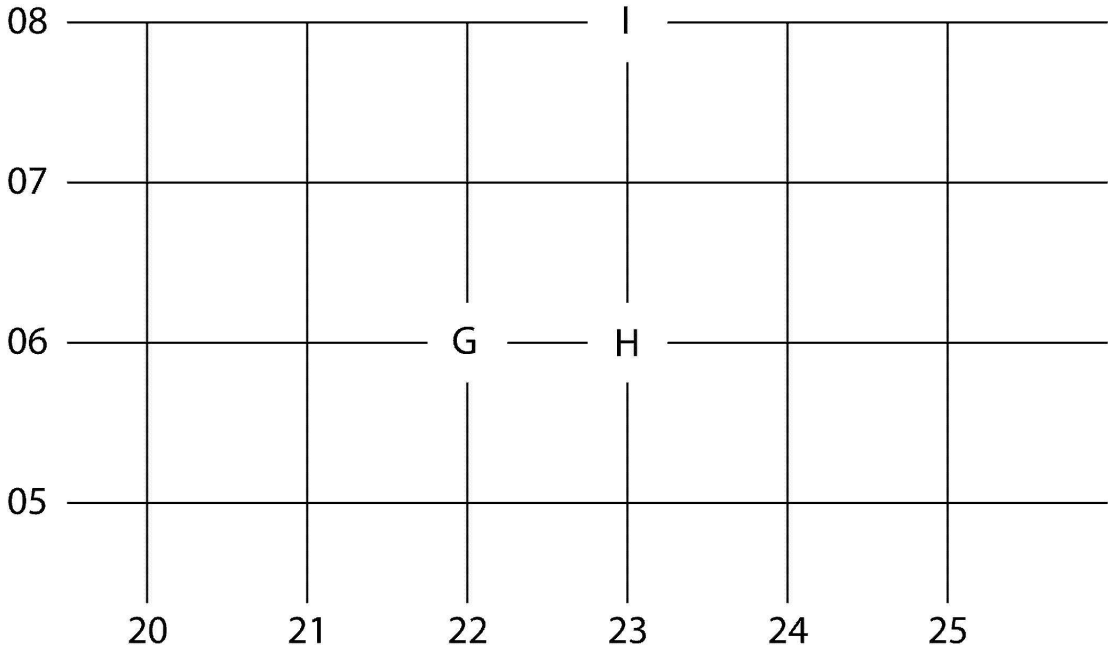


E



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8 (Q18) This diagram shows the positions of three ornithologists on a map. The squares on the map are 1000m along each side.



The ornithologists are trying to find an injured falcon which is carrying a radio transmitter. Each ornithologist has a radio tracking device. The direction-finding part of the trackers is not working but the signal strength gives an indication of the distance to the bird.

The ornithologists have the following information about the distance from themselves to the bird.

Gordon is at position 22,06; signal strength indicates more than 500m and less than 2000m.
Helen is at position 23,06; signal strength indicates more than 100m and less than 2000m.
Ingrid is at position 23,08; signal strength indicates more than 1100m and less than 1800m.

At which one of the following grid points could the bird be located?

A 22,07

B 21,06

C 22,05

D 24,07

E 23,07

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9 (Q19) The table below shows the cumulative percentage distribution of salt consumption in England in 2004.

Percentage distribution of estimated salt intake (g/day)		
	Men	Women
3 g or less	1	3
6 g or less	11	30
9 g or less	44	73
12 g or less	68	93
15 g or less	92	99
18 g or less	96	100

Assuming equal numbers of men and women in the population, what percentage of all people consumed above the recommended level of 6g per day? (Give your answer to the nearest 1%)

A 20 %

B 59 %

C 70 %

D 80 %

E 89 %

10 (Q20) A software manufacturer wants to fit some computer programs on to four 2GB discs. He reorders the programs in order of decreasing size and places the next programs to be saved in the first available disc that can take that program.

Which one of the following uses the same method as that above?

A A plumber is using lengths of pipe 12 feet long and wishes to cut some lengths. He looks at the combination of lengths to add up to 12 feet and cuts these lengths first. For the remainder he cuts the next length from the first available pipe that is long enough.

B A plumber is using lengths of pipe 12 feet long and wishes to cut some lengths. He reorders the lengths in descending order and cuts the next length from the first available pipe that is long enough.

C A plumber is using lengths of pipe 12 feet long and wishes to cut some lengths. He takes the lengths in the order listed and cuts the next length from the first available pipe that is long enough.

D A plumber is using lengths of pipe 12 feet long and wishes to cut some lengths. He alternately cuts the longest and shortest pieces still needed from the first available pipe which is long enough.

E A plumber is using lengths of pipe 12 feet long and wishes to cut some lengths. He reorders the lengths in ascending order and cuts the next length from the first available pipe that is long enough.

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11 (Q24) Alan bought a silver jug at an auction for £80. A year later, the price of antique silver had risen and he decided to sell it at auction, for which he received £100. He immediately regretted his decision and approached the buyer, who sold it back to him but he had to pay £110. Another year later, he needed some money so sold the jug to a dealer for £120.

How much profit did he make on the jug overall?

A £10

B £20

C £30

D £40

E £50

12 (Q25) An airplane flight crew starts its day in Rome and does two round trips to London in the day. On each arrival at an airport they take the next scheduled flight back. The timetable is shown below (all times are local):

Rome – London		London - Rome	
Depart	Arrive	Depart	Arrive
09:05	09:55	10:30	13:45
12:05	12:55	13:30	16:45
15:05	15:55	16:30	19:45
18:05	18:55	19:30	22:45

How long is it from take-off on their first flight to landing on their last flight of the day?

A 4h 40min

B 7h 40min

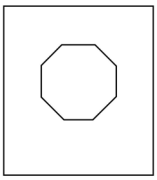
C 8h 10min

D 10h 40min

E 13h 40min

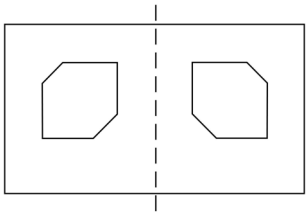
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13 (Q26) A sheet of paper has two holes cut from it. When it is folded in half the hole shown in the diagram below is seen.

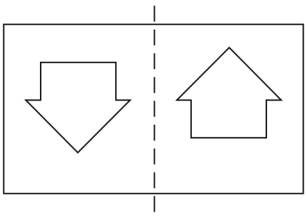


Which one of the following could **not** be the unfolded sheet of paper? (The dotted line indicates where it was folded.)

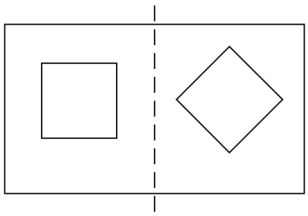
A



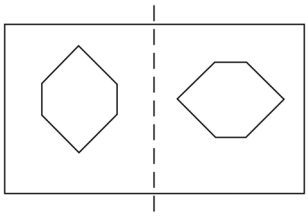
B



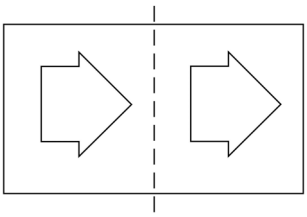
C



D



E



14 (Q30) Today I am going to treat my lawn with Greatgrass liquid lawn food. The instructions on the bottle tell me to dilute 1 part Greatgrass with 15 parts water, then apply evenly. Three hours later the treatment should be repeated, but with a mixture of 1 part Greatgrass and 24 parts water. I have worked out that I need to make up a total of 12 litres of diluted liquid for each application.

How much Greatgrass will I use altogether today?

A 960ml

B 1230ml

C 1300ml

D 1500ml

E 2460ml

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15 (Q31) The mock examination and actual GCSE results for English in one year at Morgan College are shown below:

		Actual GCSE exams	
		Grades A-C	Other Grades
Mock GCSE exams	Grades A-C	90	25
	Other grades	25	60

What percentage of pupils had their GCSE results correctly predicted by their mock examination results?

A 25.0 %

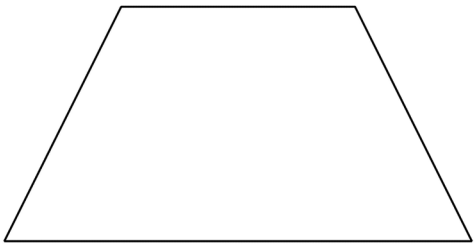
B 50.0 %

C 57.5 %

D 60.0 %

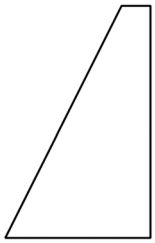
E 75.0 %

16 (Q32) The trapezium below may be cut into four identical pieces.

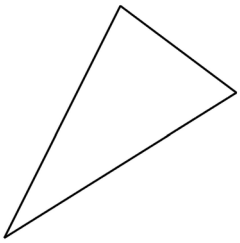


Which of the following pieces can be used four times (flips and rotations are allowed) to make up the trapezium?

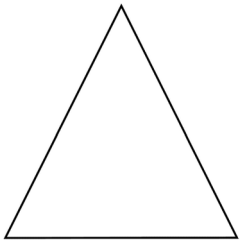
A



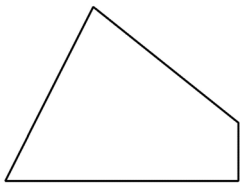
B



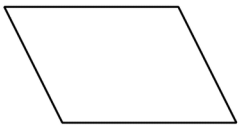
C



D



E



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17 (Q36) When Anton joined the company that he works for, 8 years ago, his salary was £29000. Each of his 8 annual salary increases has been 10% of his previous salary rounded **up** to the next multiple of £500.

What is Anton's current salary?

A £52500

B £58500

C £64500

D £67000

E £71000

18 (Q37) The Ruritanian Go Kart Drivers' Championship is decided over a series of 9 races. For each event the winner is allocated 10 points, second gets 8 points and third gets 5 points. All other finishers are given 1 point. The championship leaders' placings are shown below and the drivers all finished in each race.

Running Points Total and Placings

	Damon		Nigel		Michael		David		James	
Circuit	**Total**	**Pl.**	**Total**	**Pl.**	**Total**	**Pl.**	**Total**	**Pl.**	**Total**	**Pl.**
Rague	10	1	5	3	8	2	1		1	
Kolm	11		6		16	2	11	1	6	3
Ostrov	19	2	11	3	17		12		16	1
Vostok	29	1	19	2	18		13		21	3
Billard	39	1	27	2	23	3	14		22	
Menz	47	2	28		24		24	1	27	3
Fondra	48		38	1	32	2	25		32	3
Hahn	49		43	3	40	2	35	1	33	
Limberg	50		51	2	41		45	1	38	3

Which was the earliest race after which one of the leaders knew he could not win the Drivers' Championship?

A Billard

B Menz

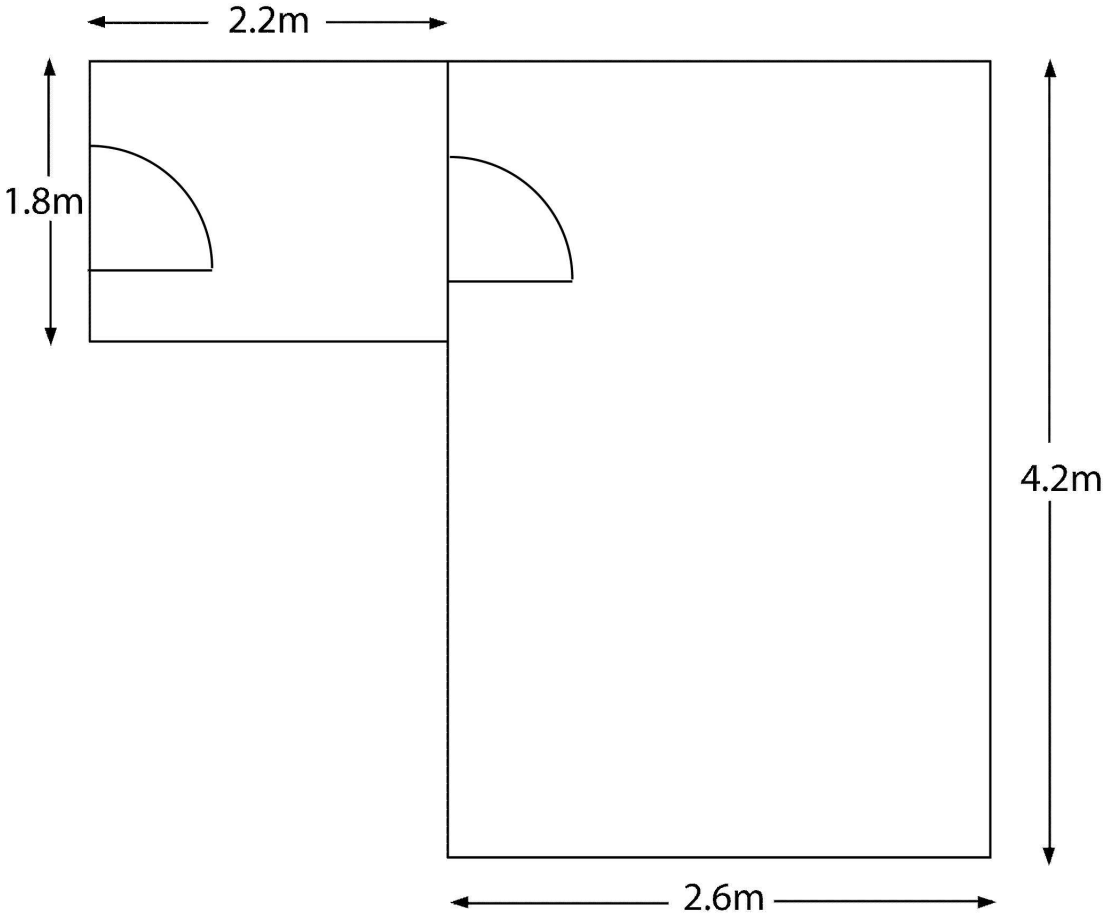
C Fondra

D Hahn

E Limberg

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19 (Q38) The diagram shows the dimensions of an office and entrance lobby. I wish to carpet them. Each must have a single piece of carpet with no joins, but I can use two separate pieces for the two rooms. As I intend to do it as inexpensively as possible, I will buy an offcut.



Which of the following offcut pieces will be suitable?

A 4.5 x 3.7 m

B 5.0 x 4.0 m

C 4.2 x 4.4 m

D 2.6 x 5.8 m

E 4.8 x 4.0 m

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20 (Q42) The Tickton town hall clock is rather a curiosity. It shows the correct time every hour on the hour, but the minute hand travels three times as fast down to 6 as it does back up to 12, though within each half of its revolution its speed is constant.

What is the correct time when the Tickton town hall clock shows quarter to the hour?

A 7.5 minutes to the hour

B 10 minutes to the hour

C 20 minutes to the hour

D 22.5 minutes to the hour

E 25 minutes to the hour

21 (Q43) The table below shows the times of sunrise and sunset and moonrise and moonset for certain towns on the same day (in hours:minutes).

	Sunrise	Sunset	Moonrise	Moonset
Avonmouth	06:57	19:09	16:25	00:56
Dun Laoghaire	07:10	19:22	16:45	01:02
Greenock	07:05	19:18	16:49	00:46
Hull	06:47	18:59	16:22	00:36
Leith	06:58	19:11	16:43	00:39
Liverpool	06:58	19:10	16:32	00:49
London	06:45	18:56	16:12	00:42

In which of the towns are the sun and moon in the sky together for the shortest time?

A Avonmouth

B Dun Laoghaire

C Greenock

D Leith

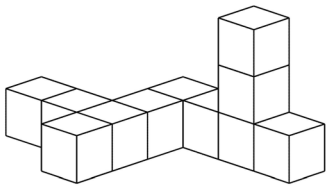
E London

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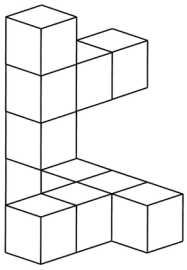
22 (Q44) Five pictures of solids made by sticking cubes together are shown below.

Which one of the following (A-E) is a different shape from **all** the others?

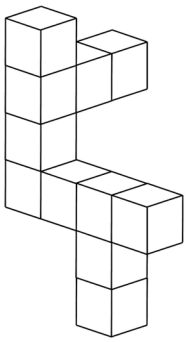
A



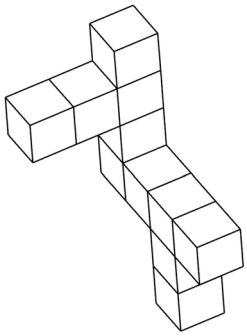
B



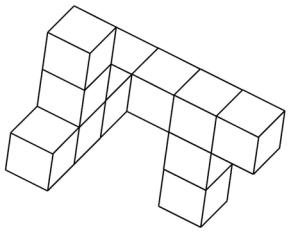
C



D



E



23 (Q48) In the elections for the mayor of Bitton, the single transferable vote system is used. In this, each voter places a '1' beside the candidate they most want and a '2' beside their second choice. Voters may only vote for two candidates. In the first round of counting all the '1' votes are counted. If no candidate has over 50% of the votes, the bottom candidate in the poll drops out and the '2' votes of all those who voted for this candidate are added to the appropriate other candidates. This process continues until one candidate has over 50% of the vote - this person is then elected mayor.

The votes after the first round of counting were as follows:

Malcolm Green	87
Susan Johnson	63
Pedro Gonzalez	45
Ian McBride	36
Wayne Daniels	18
Reeta Mathur	17
Total votes cast	254

How many candidates can still win?

A 1

B 2

C 3

D 4

E 5

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24 (Q49) Standard 100W light bulbs cost 40p each and are expected to last for 2 years in average use. Low power equivalent brightness bulbs cost £10 each, are expected to last for 8 years and use only 20W. A 100W lightbulb operated for 10 hours will use 7.5p of electricity. A 20W bulb will operate for 50 hours on the same amount of electricity. I have a light in my living room which is on for about 20 hours per week. Next time the living room bulb breaks, I will buy a low wattage bulb instead of a standard one.

How long will it be before I have recovered the extra purchase cost?

A 70 weeks

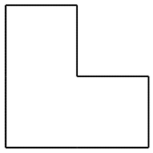
B 77 weeks

C 80 weeks

D 83 weeks

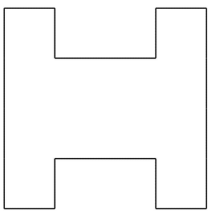
E 96 weeks

25 (Q50) In the committee room of my cricket club there are four tables, all identical to the one shown below.

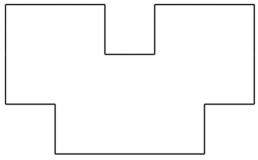


Which of the following arrangements is it **not** possible to make by putting four of these tables together?

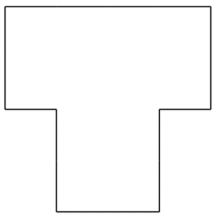
A



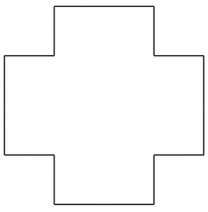
B



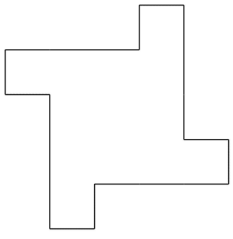
C



D



E



END OF TEST

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